

# Post Human HCI: thinking time differently

Seminar in the context of Professor Giuliana Vitiello's courses



INTERACTIVE  
TECHNOLOGIES  
INSTITUTE

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UNIVERSITÀ DEGLI STUDI  
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- Professor of Design Thinking and HCI at IST, U. Lisbon. Visiting appointment at HCII, CMU. Visiting professor, Uni As.
- Background: Fine Arts (Bachelor), Multimedia Systems (MSc), Interactive Storytelling (PhD), Aggregation (Digital Media).
- In this lecture, I will foreground the latest trends in HCI and Design theory, which grapple with how to move beyond Anthropocene concerns toward a better, more sustainable, and caring future.
- We're talking about Post-Human HCI



# DESIGN THINKING and Design -oriented approaches in HCI

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- **Design-oriented approaches in HCI** are creative ways of **solving complex problems**. Design thinking is particularly suitable for questioning conventional ways of thinking and doing, and exploring new potentials and possibilities, often neglected by traditional problem-solving approaches.
- **Design thinking** is often associated with **divergent thinking**. Design-oriented approaches offer ways to find **solutions to complex problems (hyperobjects)** such as Migrations, Sustainability, and climate change, for example.
- **Design thinking** demands **spaces where it is possible to stay with different, sometimes diverging meanings, tensions, and complexities** – dealing with multiple dilemmas, stakeholders and interests.

# The raise of INTERPRETIVE RESEARCH and HUMANISTIC HCI

Many who have done humanistic HCI have written as if they were lone wolves. Yet what we are seeing is **the emergence of a community of humanistic HCI practice\***.

**Methods reporting for qualitative and qualitative interpretive research need different methods and different reviewing standards.** This is to be expected as the community continues to grow and engagement with a broad range of disciplinary and theoretical perspectives matures#.

INTERACTIONS magazine,

\* March + April 2016, Page: 20, HUMANISTIC HCI, Jeffrey Bardzell, Shaowen Bardzell

# January + February 2024, Page: 38, INTERPRETIVE RESEARCH, Robert Soden, Austin Toombs, Michaelanne Thomas

## SEMINAR:

### Sketching More-than-human Times:

Rethinking Temporal Representations through More-than-Human, Entangled and Deep Time

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**Temporal representations can change how we see the world and design technologies.** In this seminar, we critique classic representations of time that have guided thinking in Western industrialised societies, moving towards the multiple ways time can be seen as created and creating more-than-human worlds.

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By embodying, imagining and sketching expressions of more-than-human times, we invite participants to challenge current HCI approaches to time.

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Moreover, we **propose a practical tool and design space** for HCI and design researchers **to structure dialogues and understanding of time and temporalities in more inclusive and expansive ways**

# AGENDA

## DAY 1 : The timing is tentative- lets see how it goes

First hour: Post-human HCI and thinking time differently

Present and framing of the Brief and the Mth temporalities booklet

Second hour: Start on the booklet – collage, annotation, critical unmaking of time  
The temporalities booklet can be continued during the rest of the day

## DAY 2: The timing assigned to each activity might change depending on how many groups we have and how far you went with the booklet

Feedback on the booklet is very welcome – please share it with us in which ever from you like

**1.30 h:** Based on your first-person critical reflections, Design together - Groups work together to design a new time machine inspired by the field work

**55 min:** Groups' Presentations of the ideas



# Post Human HCI and the Post Anthropocene

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In this seminar, **Post Human & Post Anthropocene HCI** is highlighted as an approach to **guide action and change into a sustainable, caring, and more fair future**, as opposed to the anxiety-generating current crisis-inspired discourse (climate–nature crisis, Pandemic crisis, migration crisis, war crisis, etc).

Through feminist post-anthropocene HCI, we can look forward to **alternatives to the *gloom and doom* picturing**— and a way out of the shame and guilt this discourse generates, to stimulate awareness that leads to action, through caring.

**The Anthropocene** is a term coined by American biologist Eugene Stoermer and Dutch geochemist **Paul Crutzen** that describes a **new geological epoch dominated by human activities at all scales**, particularly the transformation of the planet's atmosphere due to fossil-fuel burning.

Davidová, M., & Zavoleas, Y. Post-Anthropocene: The Design after the Human Centered Design Age. CAADRIA Proceedings. <https://doi.org/10.52842/CONF.CAADRIA.2020.2.203>

# Dealing with Differences Differently.

Extending HCI beyond the traditional notion of System Design- and problem-solving approaches- will **help us deal with things differently...**

Moving away from problem-solving and instrumental approaches, to become **more attuned to tensions, dilemmas, and complexities** concerning the design of (yet-to-come) digital artefacts and materials in everyday life.

HCI needs to engage with things from a **critical perspective**, exploring a new kind of design space:

*“Our task is to make trouble-stir up potent responses to devastating events, as well as to settle troubled waters and rebuild quiet places (or habitats) “ – Haraway, “Staying with the Trouble”, 2016*

# Post-Human concepts and vocabulary

## 1. The Contact Zone

*Figure developed by Donna Haraway in 2008, in the text: “When species meet”.*

**The contact zone signifies substituting subjects by their relationships:**

co-presence, interaction, and criticising asymmetrical power relationships.

This concept has already been used extensively in sociology, in the context of culture-making, when relationships are unequal, as a means to understand and trouble notions of divisions and distinction and bordering.

# The **Contact Zone**: a meta space based on feminist theories

- The contact zone encourages **clashing and grappling with different interests, belief systems and relations, and asymmetrical power relationships** - within the contact zone, no person or power overrules another.
- In the contact zone, **dilemmas take place**- regarding social justice, climate change, gender equality, urban planning, health, and ageing, to name a few...
- **Collaboration across disciplines** is imperative. **Situated knowledge** and **partial perspective** rule (Haraway, 1988).



# Some Concepts (figurations) at work in the Contact Zone:

## 2. Diffraction

To do science through a diffracted lens is to grasp different directions and intentions within a phenomenon and have the openness to be touched by the difference.

- Diffraction highlights the existence of heterogeneous and parallel meanings.
- By employing diffracting thinking within the contact zone, we commit to understanding differences within phenomena –how difference matters and for whom, like *waves in a lake coming from different directions overlapping and interfering* with one another.

Haraway writes: 'Diffraction is the production of difference, different patterns in the world, not just of the same, reflected and displaced elsewhere'.

Haraway, 1997. *Modest\_Witness@Second\_Millennium. FemaleMan@\_Meets\_OncoMouse™: Feminism and Technoscience*. New York: Routledge.

### 3. Agential Realism and Agential Cuts

Coined by Physicist Karen Barad, post-humanist agential realism starts from the premise that it is **impossible to separate the epistemological, ontological, ethical and political**.

**Agential cuts necessarily exclude some aspects and include others:**  
inside/outside are undone.

Objects or beings aren't exactly the separate, objectified, bordered, continuous-through-time-space bodies that we so constantly refer to.

**Constant exchanges and transformations are happening;** forces shift, electrons jump, and particles are entangled with particles in other spaces, without any currently measurable link or force through space.

## 4. Intra-action (contrasting interaction):

Knowledge production is reality formation

Another companion of the contact zone is the notion of **intra-action or intra-activism**- contrasting with the notion of interaction (often foregrounded in HCI research).

Through the notion of intra-action or intra-activism, Barad emphasises the **shift from representational views of things, to performative, lived and situated ones**- acknowledging that knowledge production is the formation of reality through a performative manner.

In this way, **researchers and designers take deep responsibility for actions made possible by their doings.**

## 5. Thinking with... : The relational way of thinking

To think with ... is to think of becoming together with others and being concerned about the consequences of the relationships.

- To **think with others** is about establishing positions and sensibilities to hold multiple ends and divergent positions alive.
- The **contact zone is the place** where it is possible to play with various forms of engagement, involvement and care, in search of new sensibilities.
- **Embracing the other is a central theme of the contact zone** –

A mode of thinking and researching where one is in touch and exploring differences.

# More than Human (MtH) and decentering the human

- Posthuman theory is concerned with deconstructing normative representations of 'the human', and opposing the hierarchical organisation of our ecosystem that has humans at the centre \*.
- As a concept that stems from the posthuman, **the More-than-Human is concerned with human conditions entangled with other living beings, materiality, technology and the environment.**
- In HCI, investigating such **entanglements** has come into focus ” and this has been proposed as **constituting a 'fourth wave' of HCI.**

\*[Braidotti and Hlavajova, 2018].

“[ Frauenberger, 2019; Wakkary, 2021]

# Fluid and Polyphonic Assemblages

1. Contemporary digital things are more like **fluid assemblages** than stable objects, composed of a variety of components and connections, including resources provided by various platforms and other kinds of infrastructure. They change over time and in response to activities and other inputs they are configured to register.

(Heather Wiltse: <https://medium.com/designing-fluid-assemblages>).

2. "**Polyphonic assemblage**" refers to a system or environment where diverse, independent elements (like human and non-human entities, sounds, or ideas) interact and coexist, forming a complex, multilayered whole. The concept highlights the dynamic, interconnected, and often precarious relationships within these "assemblages," emphasising collective creation, care, and the integration of more-than-human perspectives rather than human-centric autonomy.

(Anna Tsing: the Mashroom at the end of the world: <https://www.jstor.org/stable/j.ctvc77bcc>).

# 4<sup>th</sup> wave HCI and the relationship to the arts

The following slides are based on:

**Material Deconstructions of Time: Posthumanist Interventions Through Media Art.** Ciobanu, P. and Fernaeus, Y.; TOCHI, (2024).

The Usability field has evolved from first-wave HCI research on tasks, usability, and productivity to investigations of experience and engagement > the third wave of HCI ”: moving from focusing on the interface to the “human experience”.

The current interest in posthuman theory in HCI, sometimes referred to as fourth-wave HCI \*, has its reflection on topics of entanglement, first-person research, relationality, and diffraction. These topics are often embraced and explored in the art scene.

“ [Bødker, 2006; Harrison et al., 2007]

\* [Bardzell et al., 2021; Frauenberger, 2019;]

# Art as an approach in HCI

- With the emergence of digital, interactive and mixed-reality art, the contrast between the **practices within the disciplines is increasingly blurred.**
- Whether an artefact or artwork is investigated from an HCI or art standpoint, it remains conceptually, socially and politically relevant.
- It involves **understanding the value of the artefact or artwork and its potential to encourage change** in a research field or community.
- **Artistic practice holds a unique position for engaging critically with complex concepts** in ways that might not be welcome or possible through more conventional research approaches.

# Why is it important to challenge current HCI approaches to time?

- Temporality and our impact on future generations are **fundamental in research aspects related to the environmental crisis**, which is a major concern within sustainable HCI.
- Time is also central in **shifting the question of sustainability** from one of compromise and “personal morality” to “**asking what regenerative and respectful co-living looks like**”
- Within this area, post-human HCI works to unpack further the complex entanglements that form between species



# HCI and time

- The investigations of time within HCI **shifted from a focus on clock time**: the time of the machine, to investigating time as an **existential concern** socially constructed and as a cultural phenomenon <sup>\*</sup>.
- **4th Wave HCI investigates TIME as a multifaceted phenomenon** <sup>\*</sup>, entangled within interactions between human and nonhuman timescales, i.e., cultural, technological and interpersonal timescales on the one hand, and astronomical, geological and environmental scales on the other.
- Moving from focusing on efficiency and progress, to understanding time as collective, entangled and More than Human, and connected to sustainability.

<sup>\*</sup> [Rapp et al., 2022].

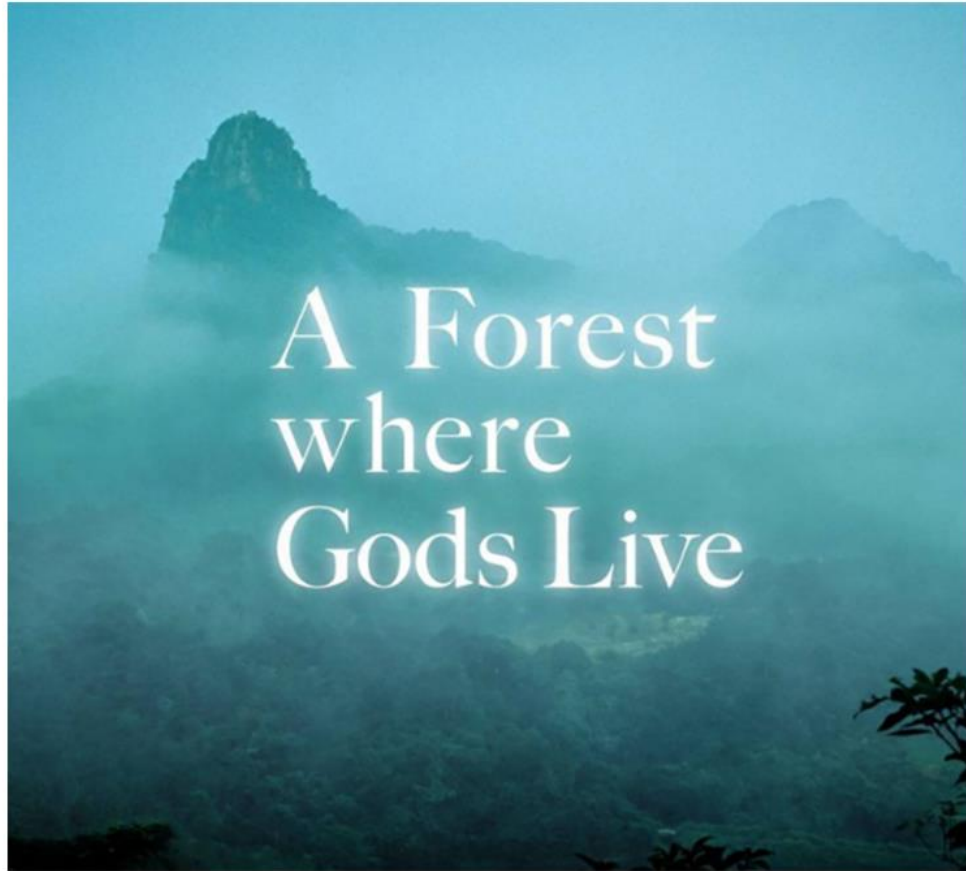
## Examples from **Material Deconstructions of Time: Posthumanist Interventions Through Media Art**. Ciobanu, P. and Fernaeus, Y.; *TOCHI* (2024).

Ciobanu et al. present **ten selected artworks**, **deconstructing established notions of time**, making space for new worldviews and emphasising entanglements and ways of cultivating relations.

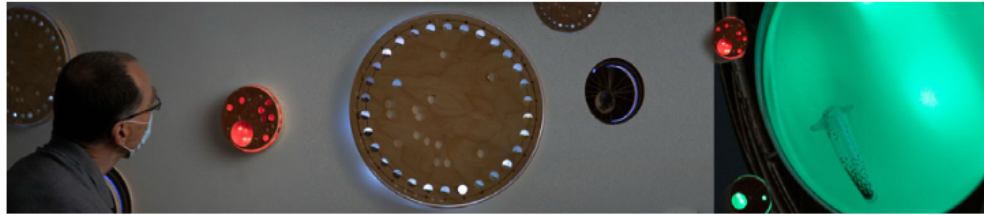
**These works may be regarded as posthumanist interventions**, as they cultivate an understanding of time and, by association, an understanding of the world from a heterogeneous and plural perspective.

Table 1. Short Description of the Media Artworks

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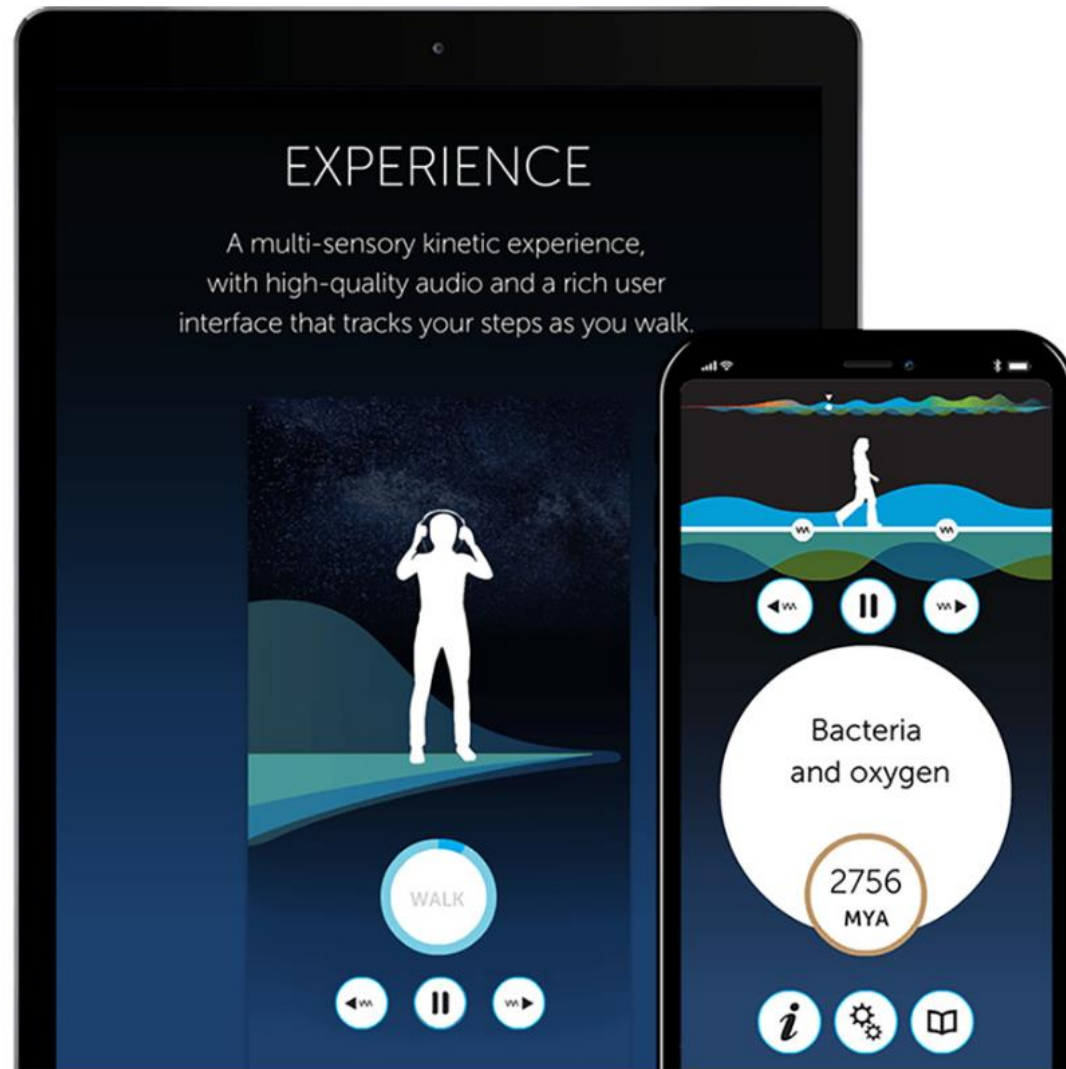


**teamLab: A Forest Where Gods Live** by teamLab is an exhibition that includes more than 20 digital works of art overlayed on various elements of Mifuneyama Rakuen Park, such as rocks, forest or mountain. Image credit: screenshot used with permission ©teamLab [teamLab, [2019a](#)]



**Allochronic Cycles** by Cesar and Lois is an installation composed of four discs that represent temporalities of the cosmos, planet Earth, Arabidopsis plant, and COVID, with an AI component that learns from these various cycles and affects them based on discrepancies between human time and ecological time. Image credit: ©Cesar & Lois [Cesar and Lois, [2021](#)]

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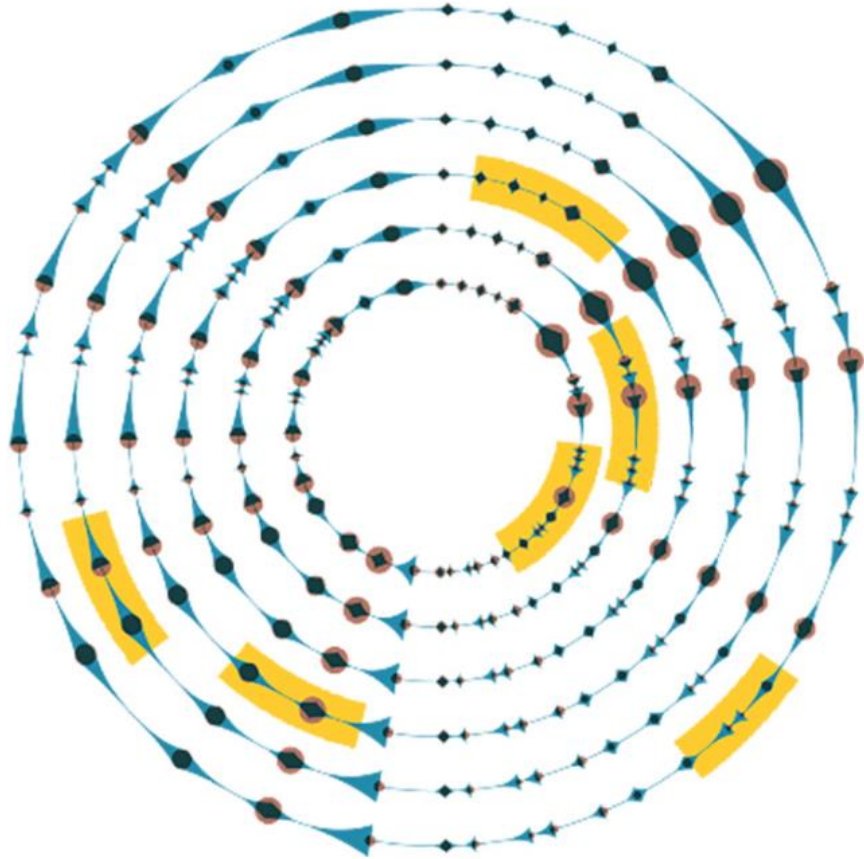


**Deep Time Walk** is a walking app created in collaboration between Robert Woodford, Geoff Ainscow, and Fred Adam. The work presents an audio narrative that guides the listener on a 4.6 km walk during which the history of the Earth is recounted. Image credit: ©Deep Time Walk CIC [Deep Time Walk C.I.C, [2017b](#)]



**Keeping Time** by Tega Brain is a visualisation of photos posted on Flickr of various plant species. The photos of a specific plant are arranged based on the timestamp, showcasing the times in a year when the plant is more photographed and thus observed. Image credit: ©Tega Brain [Brain, [2013](#)]

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**Longplayer** by Jem Finer is a musical composition scored to play for 1,000 years until it completes its cycle and starts again. Longplayer started playing at midnight, 31 December 1999, at the International Date Line, UTC +12, on the last instant of the last millennium and the first instant of the next. The artwork can be listened to online, via an app, or at various listening posts around the world. Image credit: ©Jem Finer [Finer, [1999](#)]

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**Pranatamangsa AR** by Anna Madeleine Raupach is a set of twelve black and white prints that present night sky simulations for each period in an Indonesian farming calendar. Each print has a corresponding animation that can be seen in AR, which shows the agricultural activity or event in nature occurring at that time. Image credit: ©Anna Madeleine Raupach [Raupach, [2018b](#)]

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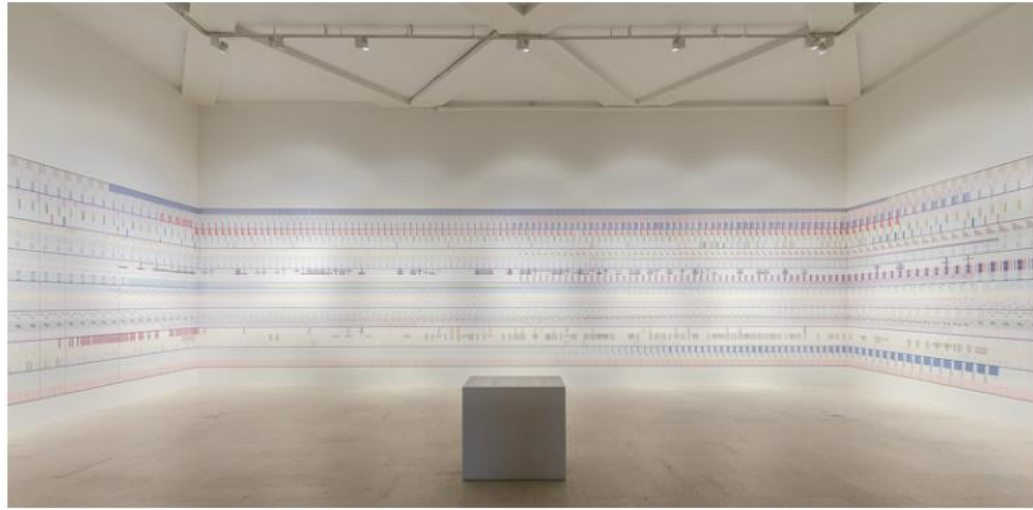
**The Phenology Clock** is an artwork created in collaboration between Natalie Jeremijenko, Jake Richardson, Blacki Migliozi, Tega Brain and others. In this project, the standard clock is reimagined through the lens of a local urban environment, with the standard clock units replaced for the months in a year, presenting how various species bud, bloom, emerge or migrate. Image credit: ©Tega Brain [Brain, 2011]

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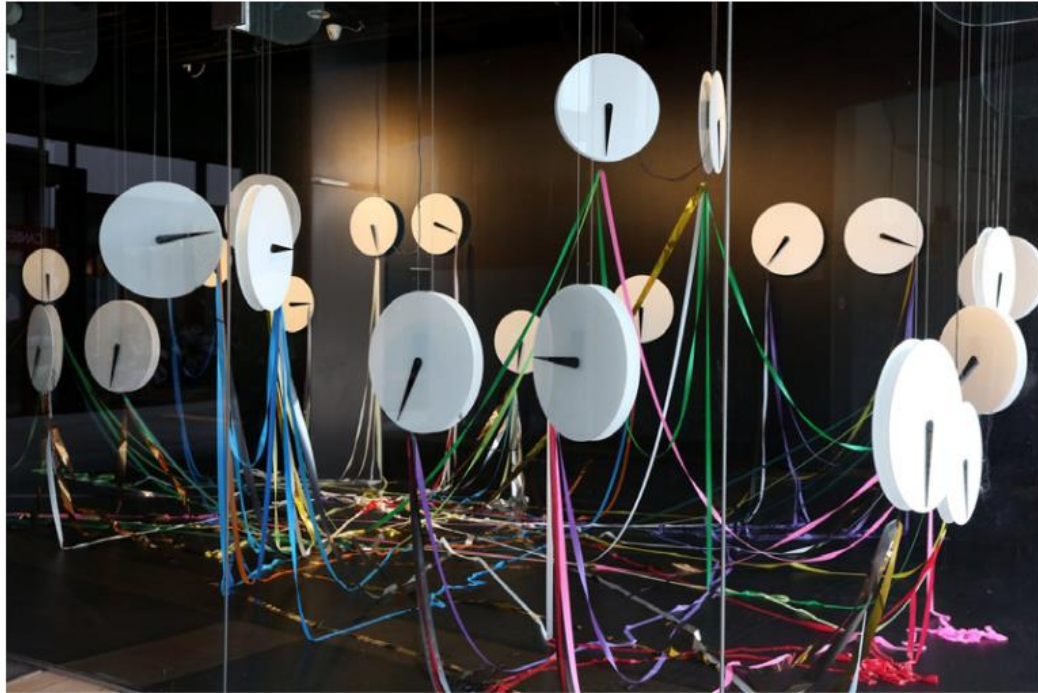
**The Present** by Scott Thrift is a collection of three different clocks that present the cycle of a day, phases of the moon, and the four seasons in a year through colour gradients. Image credit: ©The Present [The Present, [2024](#)]

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**The Room of Change** by Accurat is an installation that physically visualises data related to nature, the universe, the animal kingdom, society, hope, happiness, science and technology. The artwork visualises different events within these topics from the past, present, and future predictions. Image credit: ©Triennale Milano - photo Gianluca Di Ioia [Lupi, [2019](#)]

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**Unequal Hours** by Anna Madeleine Raupach is an installation that showcases dozens of clocks interconnected through ribbons. Each clock follows a certain timescale, such as the eclipse cycle, and is connected to associated clocks, like the clocks for the solar cycle and the moon's orbit. Image credit: ©Anna Madeleine Raupach [Raupach, [2021](#)]

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# Time interpretations and Inspirations from the artworks

The artworks materially illustrate different **understandings and views of time, deconstructing established forms of representing temporality**. However, rather than being negative, violent and destructive, the artworks represented a gentle unmaking, focusing on adding, questioning, or presenting alternatives.

1. Transitioning **from the singular perspective to a plural perspective of time**, acknowledging a diversity of ways to examine reality beyond the standards. Welcoming alternatives for specific places, uses, or cultures.

2. Highlighting the **entanglement of time as relational, social, and interconnected**, a perspective from which many projects could evolve.

3. Presenting how seemingly separate perspectives, doings, and beings **collectively create realities, opening up broader notions of what or who is included as a user or actor in a designed system**.

# Readings

## Preparation- Before the seminar:

### Entanglement HCI, the next wave

C Frauenberger.2019

[Entanglement HCI the next wave?](#)

ACM Transactions on Computer-Human Interaction (TOCHI) 27 (1), 1-27

### HCI and deep time: toward deep time design thinking.

Rahm-Skågeby, J., & Rahm, L. (2021). Human–Computer Interaction, 37(1), 15–28.

<https://doi.org/10.1080/07370024.2021.1902328>

## Inspiration - During the seminar:

### **Material Deconstructions of Time: Posthumanist Interventions Through Media Art.**

Ciobanu, P. and Fernaeus, Y. 2024. *ACM Transactions on Computer-Human Interaction*. (2024).

### Designing with more-than-human temporalities

Bessai, R., Bendor, R., and Balkenende, R. (2024), in Gray, C., Ciliotta Chehade, E., Hekkert, P ., Forlano, L., Ciuccarelli, P ., Lloyd, P . (eds.), DRS2024: Boston, 23–28

June, Boston, USA. <https://doi.org/10.21606/drs.2024.438>

### **Diagrams: Making Multispecies Temporalities Visible Open Access**

Elaine Gan, Book chapter, chapter 8,

<https://doi.org/10.1215/9781478013211-011>

Published: May 2021

### **Timing rice: an inquiry into more-than-human temporalities of the Anthropocene.**

2018, Gan, Elaine. *new formations: a journal of culture/theory/politics*, vol. 92, 2018, p. 87-101. *Project MUSE*, <https://muse.jhu.edu/article/689859>.

### **Me, the hill and my browser—investigating the role of time in posthuman interaction.**

Ciobanu, P. and Juhlin, O. 2022. Me, the hill and my browser—investigating the role of time in posthuman interaction. Nordic Human-Computer Interaction Conference (2022), 1–12.

### **Leaky Cups: Tinkering with Hydrofeminist Temporalities for HCI**

Cayla Key, Cally Gatehouse, Stevie Koepp, and Nick Taylor. 2025. Leaky Cups: Tinkering with Hydrofeminist Temporalities for HCI. In Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25). Association for Computing Machinery, New York, NY, USA, Article 408, 1–17. <https://doi.org/10.1145/3706598.3714108>

Going deeper into post-human HCI and critical temporalities:

**Characterising Interaction Design by Its Ideals: A Discipline in Transition**

Kristina Höök and Jonas Löwgren. 2021

she-ji--journal-of-design-economics-and-innovation

<https://doi.org/10.1016/j.sheji.2020.12.001>

**Thinking with care in human–computer interaction.**

Croon, A. (2022). Thinking with care in human–computer interaction. *Feminist Theory*, 23(2), 232-246.

<https://doi.org/10.1177/14647001221082294> (Original work published 2022)

**Temporal Design: Rethinking time in design.**

Larissa Pschetz, Michelle Bastian, *Design Studies*, Volume 56, 2018, Pages 169-184, ISSN 0142-694X,

<https://doi.org/10.1016/j.destud.2017.10.007>.

**Design for Temporal Cohabitation.**

Larissa Pschetz, Keili Koppel, and Michelle Bastian. 2024. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*. Association for Computing Machinery, New York, NY, USA, Article 553, 1–8.

<https://doi.org/10.1145/3613905.3644068>